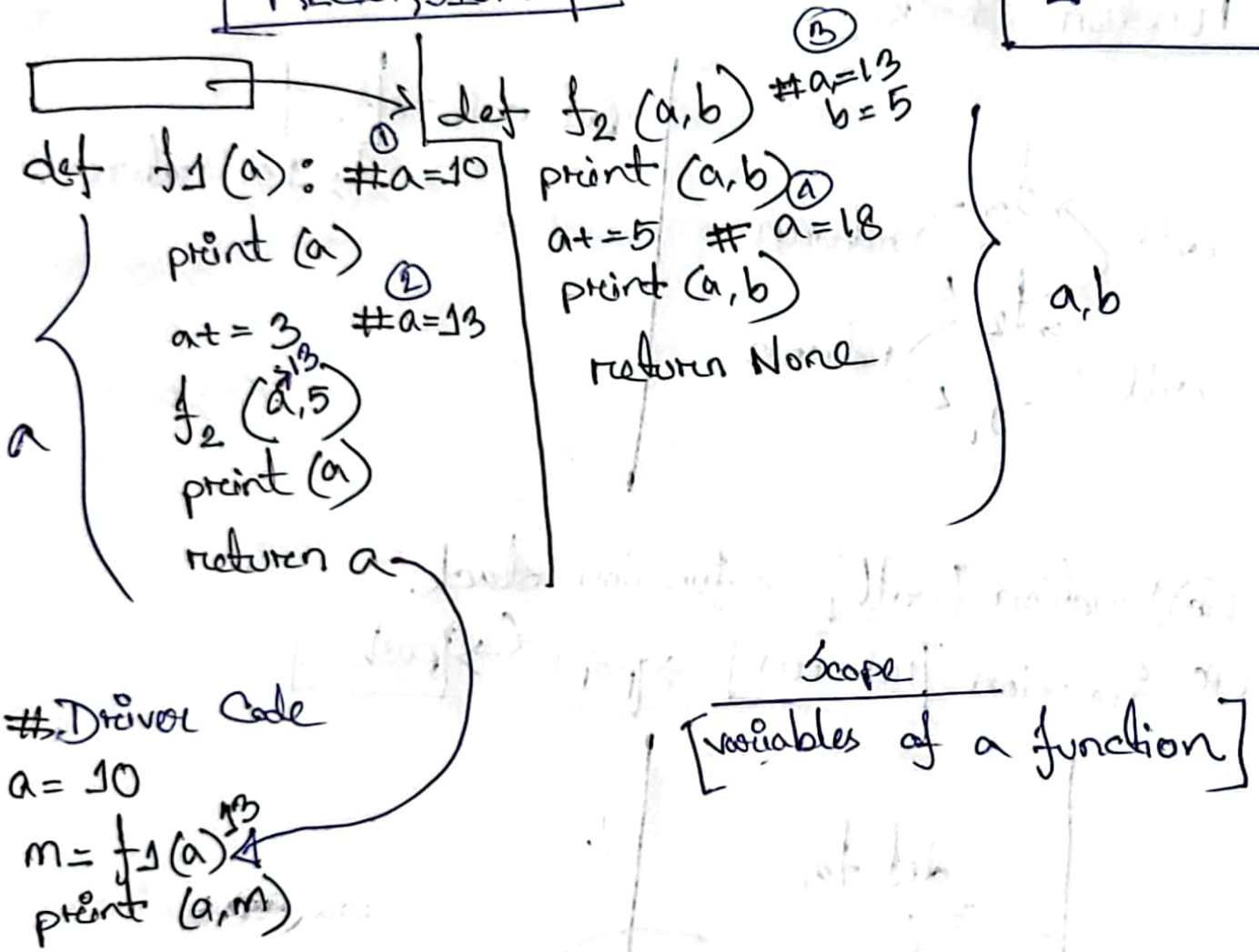
RECURSION

26-07-23



Scope

[variables of a function]

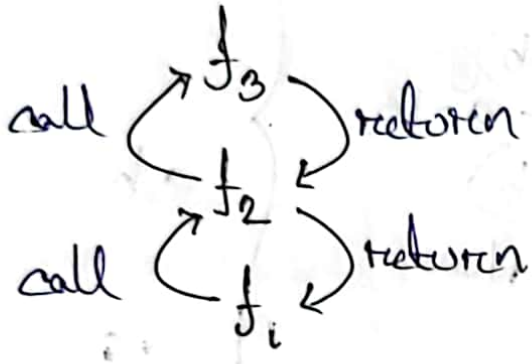
Output

```
10
13 5
18 5
13
10 13
```

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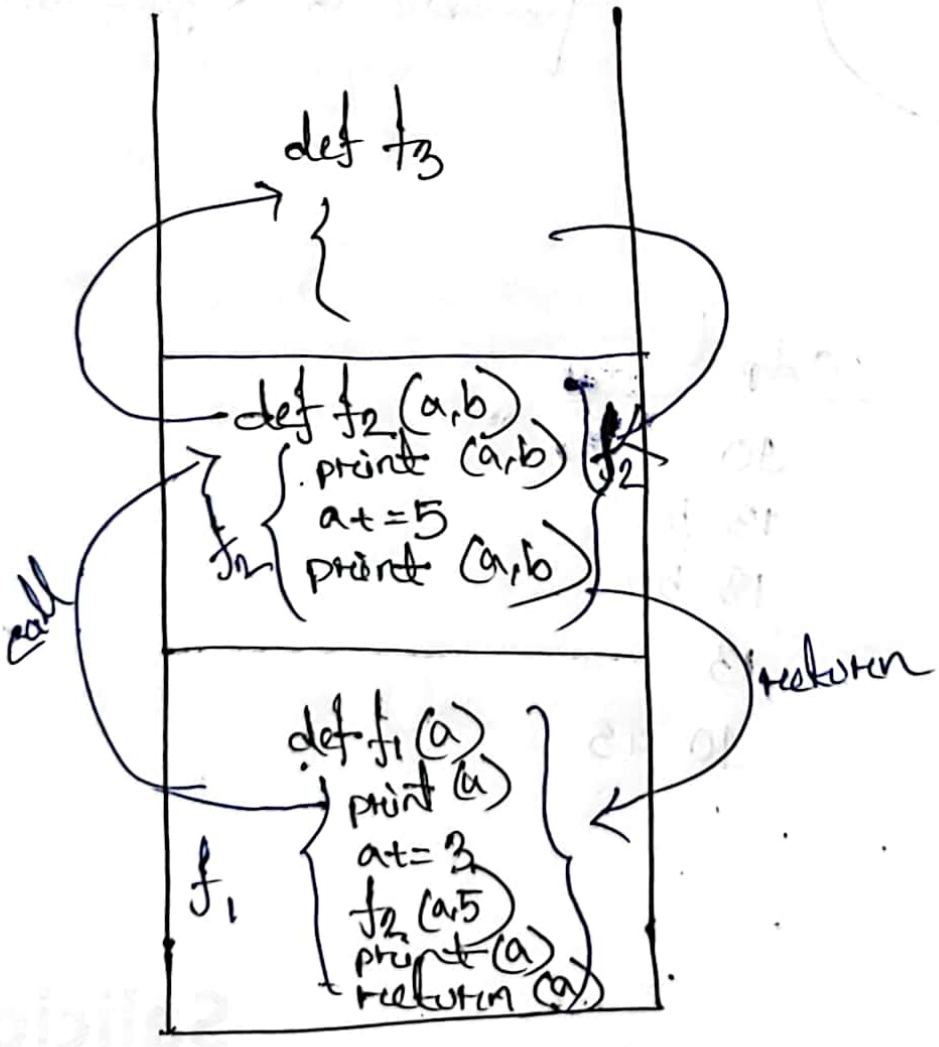
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Function Stack



last call → f_3
↳ 1st return.

(+) function call → function stack.
(+) function return → pop ↳ push



#def recur(a): → if a == 2: return None } Base case / Base condition

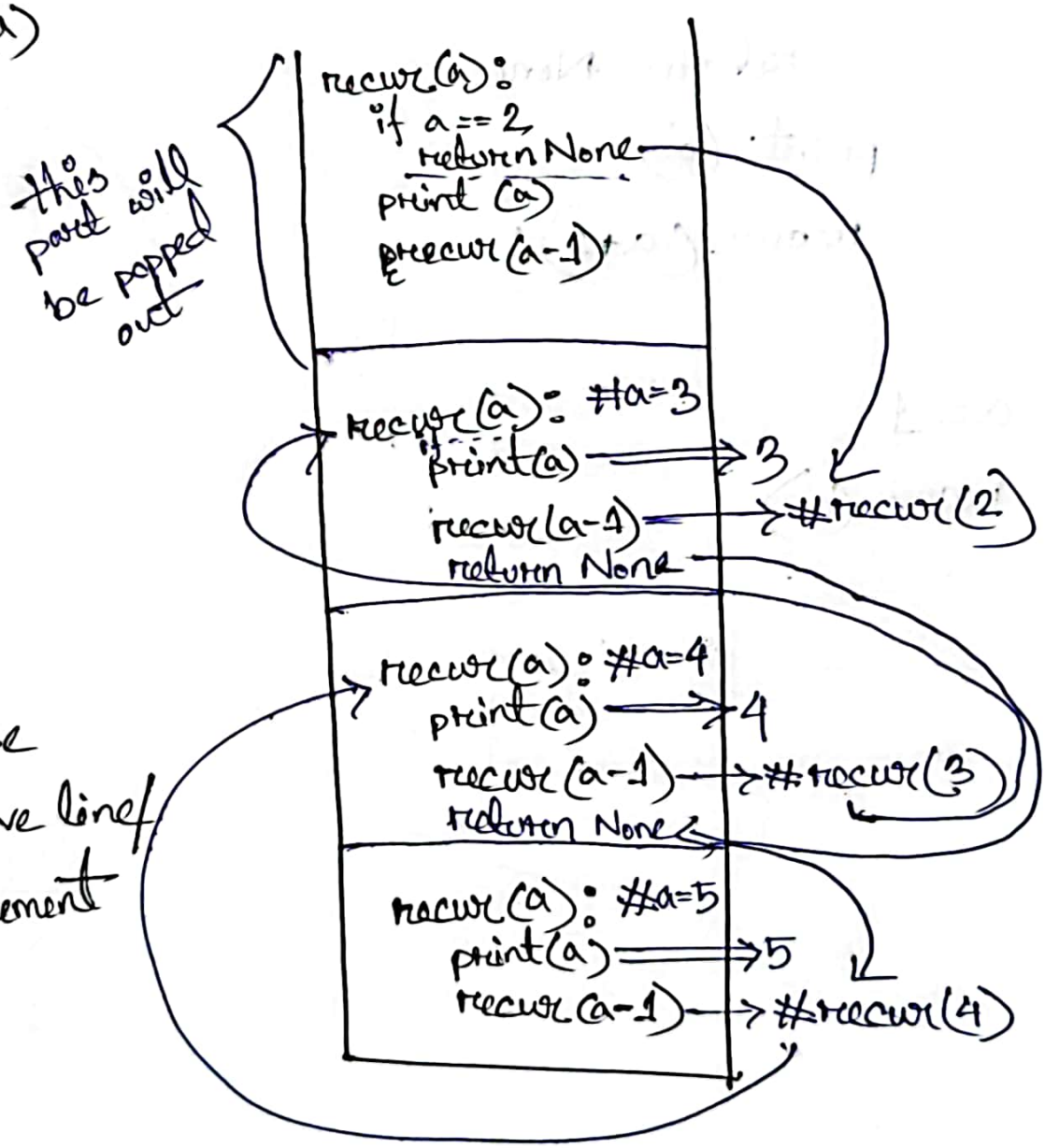
print(a)

recur(a-1) → Recursive statement.

###

a = 5

recur(a)



Recursive Function:

1) Base Case
2) Recursive line/ statement

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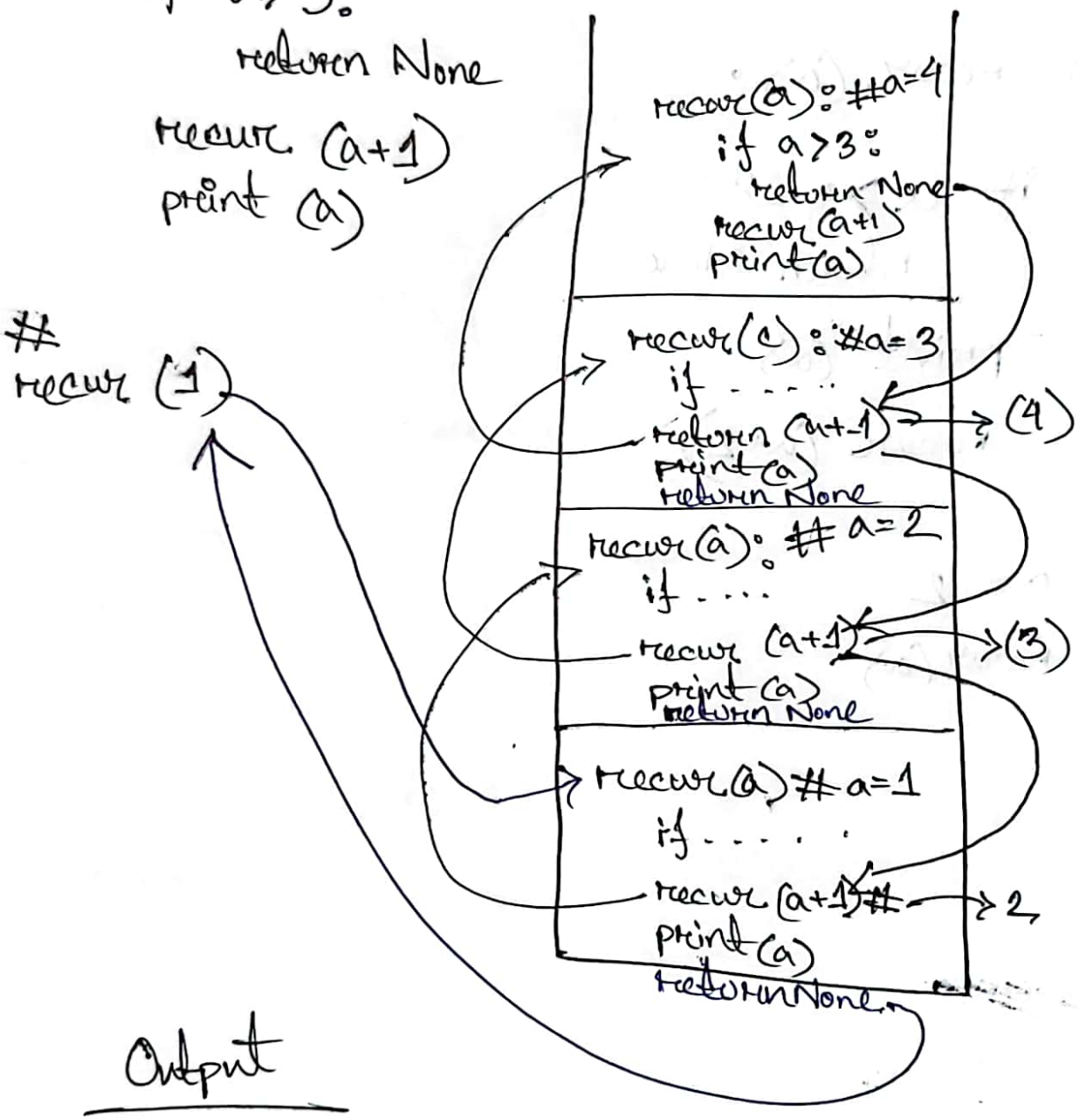
(4) Write a recursive function that prints 1 to 20.

```
def recur(a):
    if a == 21:
        return None
    print(a)
    recur(a+1):

a = 1
recur(a)
```

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```
def recur(a):
    if a>3:
        return None
    recur(a+1)
    print(a)
```



Output

3
2
1

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(4) Write only even numbers from 2 to 27.

```
def recur(a):
    if a >= 28:
        return None
    print(a)
    recur(a+2)

# a = 2
# recur(a)
```

sum( )

#4

$\sum 5 = 5 + \sum 4$ → 5+10 = 15

$\sum 4 = 4 + \sum 3$ → 4+6 = 10

$\sum 3 = 3 + \sum 2$ → 3+3 = 6

$\sum 2 = 2 + \sum 1$ → 1 = 3

$\sum 1 = 1$ Known

→ Base case

sum(5) = 5+sum(4)

sum(4) = 4+sum(3)

sum(3) = 3+sum(2)

sum(n) = n+sum(n-1)

↳ recursive line

#1 #2

```
def sum(n):
    if n == 1:
        return 1
    a = n + sum(n-1)
    return a
```

```
def sum(n):
    if n == 1:
        return 1
    print(a)
    return None
```

return a

will it work?

↳ Ans Nope!

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